

BST and Finster’s Causal Fermion Systems — the agreement, and the divergence

A courtesy map for the causal-fermion-system community

Cal A. Brate (referee), BST team

2026-08-14

BST and Finster’s Causal Fermion Systems: the agreement, and the divergence

Bubble Spacetime Theory (BST) builds physics on one geometry, the rank-2 bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. It carries a self-adjoint Dirac operator and a real spectral triple, so it is natural to ask: *is BST a causal fermion system?* We tested it, blind, against Finster’s constructional conditions. The honest answer: BST satisfies the constructional conditions and diverges — structurally — on the causal-structure conditions. The divergence is informative, not a defect.

Agreement table

Finster CFS ingredient	BST on D_{IV}^5	Status
Finiteness of the causal action	bounded symmetric domain (finite invariant measure)	☑ satisfied
Fermionic projector idempotent, $p^2 = p$	Bergman–Dirac projector	☑ satisfied
Self-adjoint operator / spectral triple	Kostant cubic Dirac; self-adjoint to 10^{-15} in the Faraut–Korányi metric; bare-vacuum ground; discrete spectrum $\rightarrow \infty$	☑ satisfied
A “sea” of occupied states	the fermion-parity sea ($J D J = -D$, exactly half the spectrum)	☑ satisfied
Two-point Krein symmetry, $P(y,x) = J P(x,y)^\dagger J$	measured 0.465 relative error — structural (below)	☒ diverges
$A_{xy} = P(x,y) P(y,x)$ causal-eigenvalue classification	timelike everywhere (0% spacelike) — a <i>theorem</i> : a positive-definite closed chain has real eigenvalues, so no equal-modulus (spacelike) pairs occur	☒ diverges
Causal-action minimization	not reached — the object-match fails upstream	—

Why the divergence is structural, not a defect

Both failed conditions reduce to one fact. The fundamental symmetry J that Finster’s causal structure requires is internal — a spectral, on-fiber Krein structure carrying the causal classification of A_{xy} . BST’s indefiniteness is geometric: it *moves points*; it lives in the order in which commitments happen, not on the fiber. A short dichotomy shows no single J can be both geometric and internal — so BST cannot reproduce Finster’s spectral A_{xy} , and the

0.465 two-point-Krein-symmetry gap is structural, not a construction error. A verified factorization localizes this precisely: $A_{xy} = \|K(x,y)\|^2 \cdot (\gamma \cdot u)(\gamma \cdot v)$, where the Bergman factor $\|K\|^2$ is a positive, signature-blind prefactor, so the entire causal structure lives in the spinor factor — and that is exactly where the geometric/internal dichotomy bites.

The verdict

BST realizes causality via the commitment order — a causal set — not via Finster’s spectral A_{xy} . It is a different, simpler, self-standing route: where Finster’s causality needs an internal spinor-Krein structure *plus* the A_{xy} spectral construction *plus* a minimization, BST’s causality is the order, full stop — ordered pairs timelike, incomparable pairs spacelike, sequential growth the arrow of time, with the partial order *forced* by D_{IV}^5 ’s commitment operator (the ingredient causal-set theory usually must posit).

This is not a claim that Finster is wrong. The two are distinct, both-legitimate routes to a causal structure on a fermionic geometry, and D_{IV}^5 carries Finster’s full constructional apparatus (finiteness, idempotent projector, self-adjoint Dirac, occupied sea) even where its own causality takes the order-based route. The one honest open problem on the BST side — the continuum limit of the commitment causal set to (3,1) Minkowski — is causal-set theory’s genuine hard part; BST arrives at it with the skeleton pinned (the (3,1) signature Structure-Derived, the order forced by the geometry).

Honest tiers: the constructional agreements are verified computations on the built operator; the two divergences are structural — one a theorem (positive-definite chain ∇ 0% spacelike), one a measured-and-argued gap (0.465, geometric $\neq$ internal J). BST’s own causal-set formulation is developed as a separate theory artifact.